1. Measures of Spread

1.1. Variance of Discrete Data

- The variance is a measure of spread.
  - It is denoted by σ^2. It can be found using the formulas below:

$$\sum \frac{(x_i - \overline{x})^2}{n}$$

$$\sum_{i=1}^{n} \frac{x_i^2 - \overline{x}^2}{n}$$

Coded Variance

- Coded data does not affect the variance.
  - This is because the actual spread of the values does not change when they are transformed.

$$\text{Variance} = \frac{\sum (x-b)^2}{n} - \left(\frac{\sum (x-b)}{n}\right)^2$$

Grouped Variance

- The variance of 2 sets of data, or grouped variance, can be found by adding the mean of squares and subtracting it using the mean squared.

Suppose a data set x exists where the sum of the data is $\sum x$ and the number of values is n_x, and another data set y exists where the sum of the data is $\sum y$ and the number of values is n_y.

$$\text{Grouped Variance} = \frac{\sum x^2 + \sum y^2}{n_x + n_y} - \left(\frac{\sum x + \sum y}{n_x + n_y}\right)^2$$

Example

The Znotes Archery club has 12 Junior members and 20 Senior members. For the Junior members, the mean age is 15.5 years and the standard deviation of the ages is 1.2 years. The ages of the Senior members are summarised by $\sum y = 910$ and $\sum y^2 = 42850$, where y is the age of a Senior member in years. Find the mean and variance of all 32 members of the club.

Answer

Let x denote the data of the junior members, such that $n_x = 12$ and $\frac{\sum x}{12} = 15.5 \implies \sum x = 12 \times 15.5 = 186$.

The combined mean of both data sets is:

$$\frac{186 + 910}{12 + 20} = 34.25$$

The standard deviation of the juniors is 1.2, meaning the variance is 1.2^2.

$$1.2^2 = \frac{\sum x^2}{12} - (15.5)^2 \implies \sum x^2 = 2900.28$$

Now we can find the combined variance:

$$\sigma^2 = \frac{2900.28 + 42850}{12 + 20} - (34.25)^2 = 256.63$$

1.2. Standard Deviation

- The standard deviation is the measure of the spread of values from the mean.
  - It is denoted by σ.
- If coded mean and sums are given, use as it is, standard deviation not altered

$$\text{Standard deviation} = \sqrt{\text{Variance}} = \sqrt{\frac{\sum (x_i - \overline{x})^2}{n}} = \sqrt{\frac{\sum x_i^2}{n} - \overline{x}^2}$$

$$\text{Standard deviation} = \sqrt{\text{Variance}} = \sqrt{\frac{\sum (x_i - \overline{x})^2 f_i}{\sum f_i}} = \sqrt{\frac{\sum x_i^2 f}{\sum f_i} - \overline{x}^2}$$

1.3. Variance in Frequency Table

$$\frac{\sum (x_i - \overline{x})^2 f_i}{\sum f_i}$$

or

$$\frac{\sum x_i^2 f_i}{\sum f_i} - \overline{x}^2$$

{W04-P06} Question 4:

The ages, x years, of 18 people attending an evening class are summarised by the following totals:

$\sum x = 745$, $\sum x^2 = 33\,951$

i. Calculate the mean and standard deviation of the ages of this group of people.

ii. One person leaves group and mean age of the remaining 17 people is exactly 41 years. Find age of the person who left and standard deviation of the ages of the remaining 17 people.

Solution:

Part (i):

$$\sigma = \sqrt{\frac{\sum x^2}{n} - \overline{x}^2} \qquad \overline{x} = \frac{\sum x}{n}$$

$$\sigma = 13.2 \qquad \overline{x} = 41.4$$

Part (ii):

The total age of the 18 people

$$\sum x = 745$$

Find the total age of the 17 people

$$\sum x = 41 \times 17 = 697$$

Subtract the two to get the age

$745 - 697 = 48\ years$

Calculating the new standard deviation

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Find the $\sum x^2$ of the 17 people

$$\sum x^2 = 33\,951 - 48^2 = 31\,647$$

Find the standard deviation

$$\sigma = \sqrt{\frac{31\,647}{17} - (41)^2} = 13.4$$

{S13-P62} Question 2:

A summary of the speeds, x kilometres per hour, of 22 cars passing a certain point gave the following information:

=$\sum (x-50) = 81.4$ and $\sum (x-50)^2 = 671.0$

Find variance of speeds and hence find the value of $\sum x^2$

Solution:

Finding the variance using coded mean

Variance $= \frac{671.0}{22} - \left(\frac{81.4}{22}\right)^2 = 16.81$

Find the actual sum

$$\sum x = 81.4 + (22 \times 50) = 1181.4$$

Put this back into variance formula

$$16.81 = \frac{\sum x^2}{n} - \left(\frac{1181.4}{22}\right)^2$$

$$\therefore \sum x^2 = 2900.5 \times 22$$

$$\sum x^2 = 63811$$

1.4. Range

- Interquartile range = Upper quartile - Lower quartile
- Range = Largest value - Smallest value

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